

ISTE.240.602

EventHub: Final Project Report

Group 9

I. Title Page

- **Project Title:** EventHub – An Event Management & Discovery System
- **Semester & Year:** Spring 2026
- **Team Members:**
 - **Ali Faisal** (ID: 415002117) – *Lead: Subscription & Administrator Modules*
 - **Syed Rayan** (ID: 764000072) – *Lead: Event & User Modules*

Rubric

FRONTEND (20 pts) – CLOS 2, 3 and 5

- |— ***18-20: Professional + Bootstrap + responsive + toasts + loading states***
- |— ***14-17: Clean + responsive + basic response messages***
- |— ***10-13: Basic CSS + not responsive***
- |— ***0-9: No CSS or broken***

REST API (25 pts) – CLOS 4 and 5

- |— ***22-25: Each student has own controller + All 6 methods***
- |— ***17-21: Each has controller + 4-5 methods***
- |— ***12-16: Only 1-2 controller + 2-3 methods***
- |— ***0-11: Only one controller + Less than 2 methods***

JPA (20 pts) – CLO 4

- |— ***17-20: Each student has own repository + 5+ methods + entity classes***
- |— ***13-16: Each has repository + 3-4 methods***
- |— ***9-12: Only 1-2 repository + 1-2 methods***
- |— ***0-8: One repository for all → ZERO for section***

GIT (10 pts) – CLO 1

- |— ***9-10: Branches + 5+ commits + good messages + PRs***
- |— ***7-8: Branches + 3-4 commits***
- |— ***5-6: One branch + 1-3 commits***
- |— ***0-4: Single commit***

DEMO (15 pts) – CLOS 2, 3 and 5

- |— 13-15: 5+ pages + all CRUD + no errors
- |— 10-12: 4-5 pages + most CRUD
- |— 7-9: 3 pages + basic CRUD
- |— 0-6: Less than 3 pages

FINAL REPORT (10 pts) – CLO 1

- |— 9-10: Complete documentation + API docs + team contributions + screenshots
- |— 7-8: Most sections present + basic API docs
- |— 5-6: Basic report + missing key sections
- |— 0-4: No report or very incomplete

FINAL PRESENTATION (10 pts) – CLO 1

- |— 9-10: Clear explanation + live demo works + answers questions confidently
- |— 7-8: Good presentation + minor demo issues
- |— 5-6: Basic presentation + demo has problems
- |— 0-4: Cannot explain code + demo fails

Critical Deductions (Applied AFTER scoring)

Violation	Deduction
Missing student names/IDs in README	-10 points
Missing student names/IDs (as comments) at the top of their own classes	-10 points
Only one repository for all entities (not per student)	-8 points (ZERO for JPA section)
Using Mustache templates instead of HTML/JS/REST APIs	-10 points
Only one controller/service for all students	-11 points
No proper database seeding (0 records) or missing/incorrect configuration	-10 points
Absence during the presentation	-20 points

II. Manifesto Page

Individual Contribution & Integrity Statement: *We, Ali Faisal and Syed Rayan, certify that this project is the result of our own academic work and collaboration.*

- **Syed Rayan** implemented the full-stack lifecycle for **Event** and **User** entities, including JPA repositories with custom JPQL, REST controllers with 6 HTTP methods, and independent service layers.
- **Ali Faisal** implemented the **Subscription** and **Administrator** modules following the same architectural standards.
- The frontend was developed using **Bootstrap 5** and the **Fetch API**, completely removing the previous Mustache engine to satisfy the Phase 2 requirements.

III. Abstract

EventHub is a comprehensive event management system that is designed to provide a seamless connection between event discovery and administration control. The project represents a shift from a traditional server-side template structure to a modern N-tier layered architecture. Built using Spring Boot 3.x, Spring Data JPA, and a MySQL database, the platform delivers persistent data management. The frontend is developed to be a dynamic , responsive standalone interface that can communicate with the backend using asynchronous RESTful APIs. This enables a smooth user experience without any page interruptions or reloads.

IV. Problem Definition & Solution Context

In today's digital world, event details are often fragmented, making it hard for users to discover opportunities and for the organizers to manage an event. EventHub solves this problem by:

- **Discovery and Responsibility:** Users can discover events and can instantly register through a specific dedicated module.
- **Administrative Supervision:** Administrators oversee the process, controlling access to the organization and maintaining data integrity.
- **Technical Framework:** The solution involves employing JPA persistence to solve the issue of temporary in-memory storage. This guarantees that every event discovered and subscription is captured onto the iste240 database permanently.

V. Brief Description of Current Similar Solutions

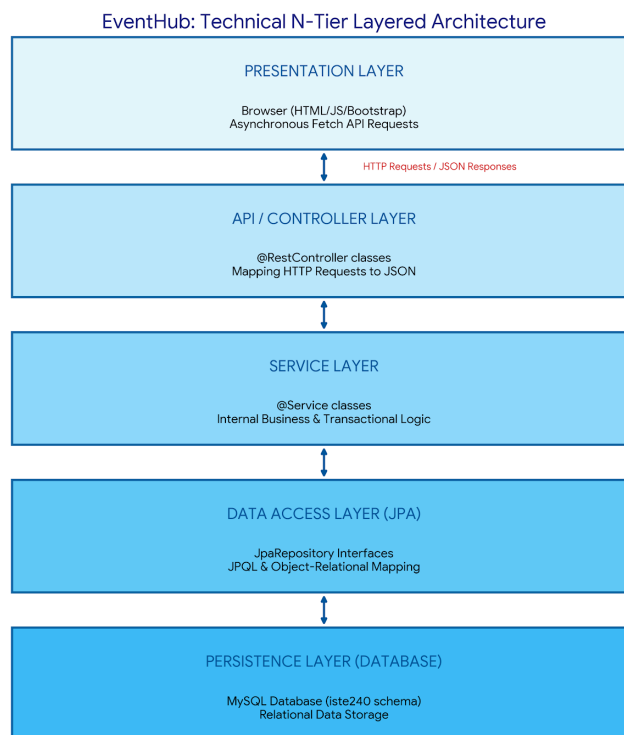
- **Eventbrite:** A top event management system that implements a similar RESTful architecture to make event discovery and registration possible
- **Meetup:** Centered around community-based discovery, using mobile-compatible frontends and decoupled backends to manage high frequency user interactions.

VI. Technologies Used

The project implements a **Decoupled Layered Architecture**:

- **Presentation Layer:** Standalone HTML/JS files that use the **Fetch API** to send and receive JSON data.
- **Controller Layer:** `@RestController` classes that map HTTP requests to service methods.
- **Service Layer:** Dedicated classes (`SubscriptionService`, etc.) that handle business logic independently.
- **Data Access Layer:** JPA Repositories that abstract SQL complexity and use **JPQL** for custom searches.

Technical Architecture



Decoupled architecture utilizing Spring Boot REST APIs and MySQL persistence.

VIII. Key Website Features

- **Real-Time CRUD:** Complete Create, Read, Update, and Delete operations for all four modules.
- **Advanced Searching:** Custom search capabilities implementing the SQL LIKE operator via JPQL to locate events by title or administrators by department.
- **Database Seeding:** Automatic system initialization via the DataLoader class to guarantee immediate out-of-the-box usability.
- **Transactional Updates:** Secure data modification powered @Modifying and @Transactional annotations.

IX. Design Summary

- **Styling:** Bootstrap 5 was implemented to design a professional, clean dashboard, with cards and striped tables integrated; this ensures data is easily scannable.
- **Interaction Scenarios:** A user inputs their email and event title into a form. Javascript intercepts the submission, then sends out a POST request to the API, the table is then updated automatically needing to refresh the page.

X. Database Design & Implementation

The database schema was structured to uphold the relational integrity within the iste240 database schema.

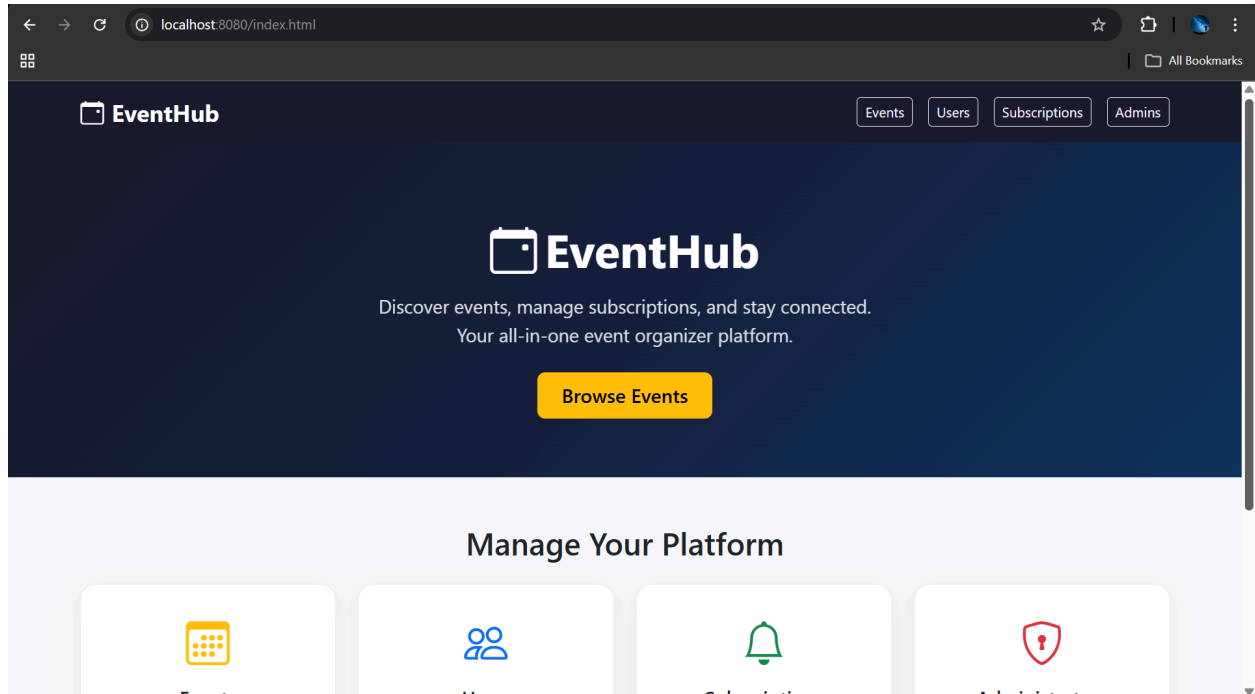
- **Implementation:** Hibernate was configured with ddl-auto=update to automatically generate tables from Java @Entity classes.
- **Performance:** Long ID auto-generation was adopted to handle scalability demands.
- **Relationships:** A Many-to-One relationship was created between Subscription and Administration to track approval workflows.

XI. GIT Repository Link

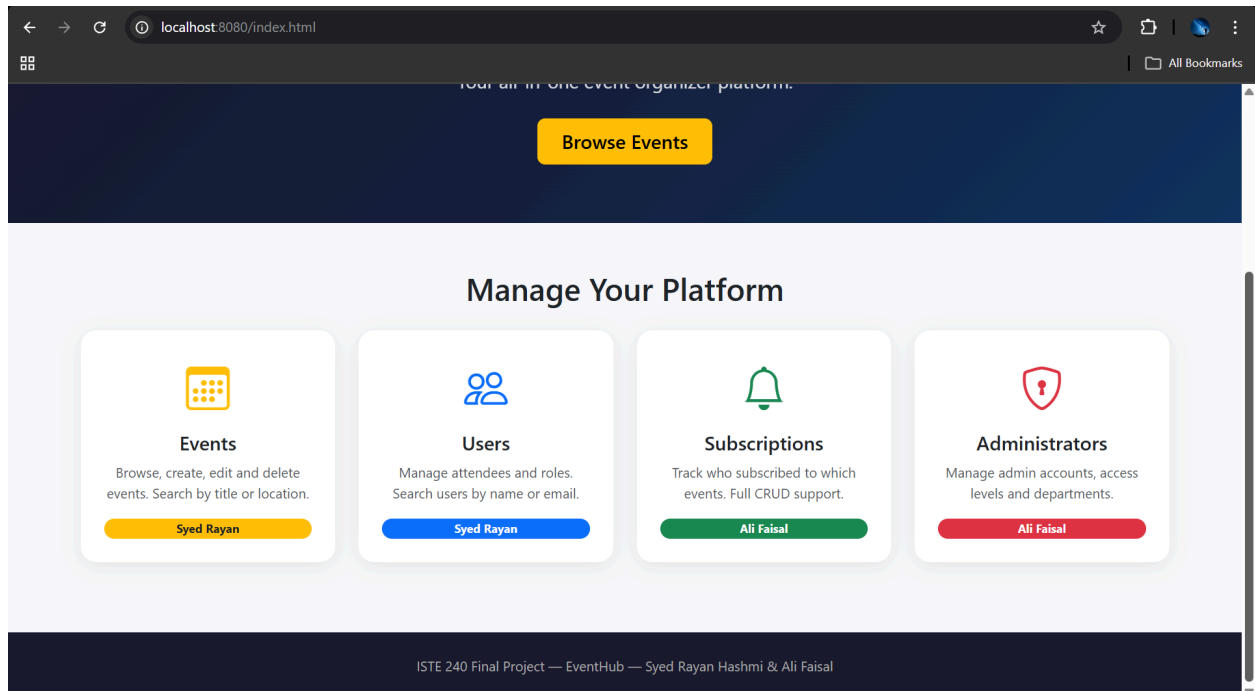
<https://github.com/SyedRayan0/ISTE240proj.git>

XII. Screenshots

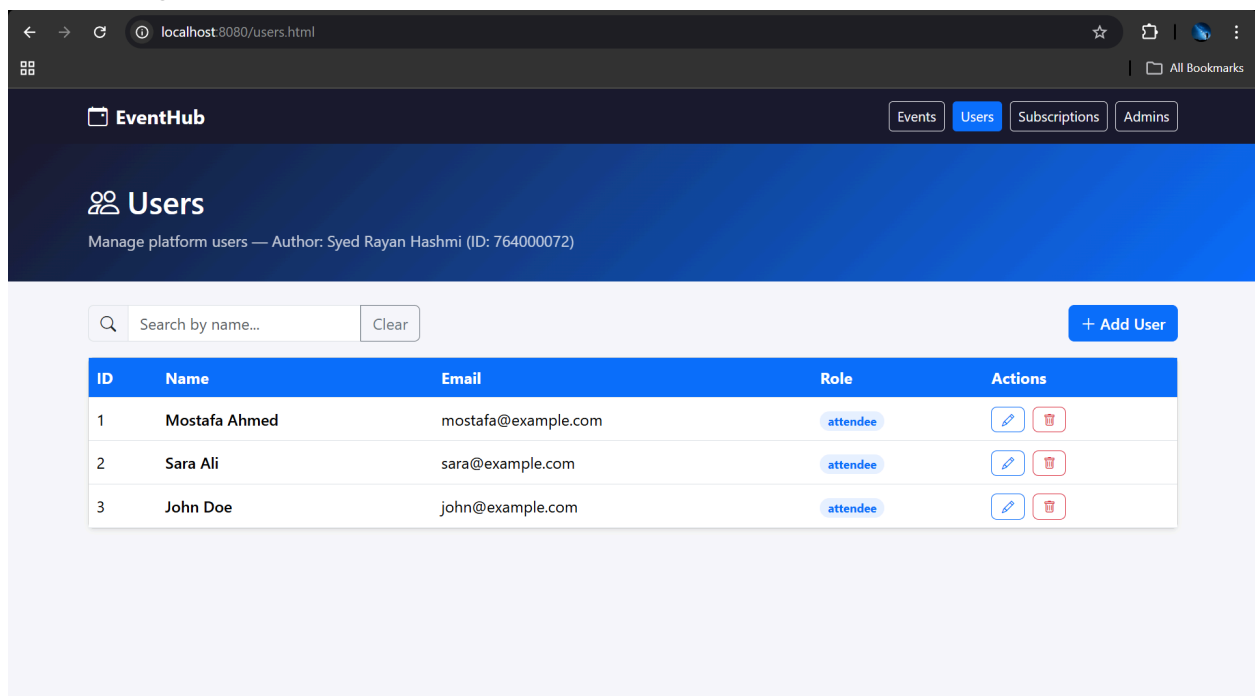
Index page pt1



Index page pt2



Users page



Events page

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localhost:8080/events.html

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All Bookmarks

EventHub

EventsUsersSubscriptionsAdmins

📅 Events

Manage all platform events — Author: Syed Rayan Hashmi (ID: 764000072)

🔍 Search by title...

Clear

+ Add Event

ID	Title	Description	Date	Location	Actions
1	Hackathon 2026	Event Description	2026-02-20	Main Hall	
2	Tech Workshop	Event Description	2026-02-20	Lab A	
3	AI Conference	Event Description	2026-02-20	Auditorium	

Subscriptions page

EventHub

EventsUsersSubscriptionsAdmins

🔔 Subscriptions

Manage event subscriptions — Author: Ali Faisal (ID: 415002117)

🔍 Search by event title...

Clear

+ Add Subscription

ID	Subscription ID	Email	Event Title	Actions
1	SUB-101	ali@rit.edu	Hackathon 2026	
2	SUB-102	student@rit.edu	Tech Workshop	
3	SUB-103	mostafa@example.com	AI Conference	

Administrators page

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localhost:8080/administrators.html

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All Bookmarks

EventHub

EventsUsersSubscriptionsAdmins

🛡️

Administrators

Manage admin accounts — Author: Ali Faisal (ID: 415002117)

🔍

Search by department...

Clear

+ Add Administrator

ID	Admin ID	Access Level	Department	Actions
1	ADM-001	Full	Operations	✎ 🗑
2	ADM-002	Read-Only	Marketing	✎ 🗑
3	ADM-003	Restricted	IT	✎ 🗑